FaceIn

Face Recognition Attendance System

Employee attendance by face recognition, checked against the workplace location.



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The Problem & Our Solution

THE CHALLENGE

Manual attendance takes time and effort
Identity can be hard to verify

Attendance may be logged from unapproved locations

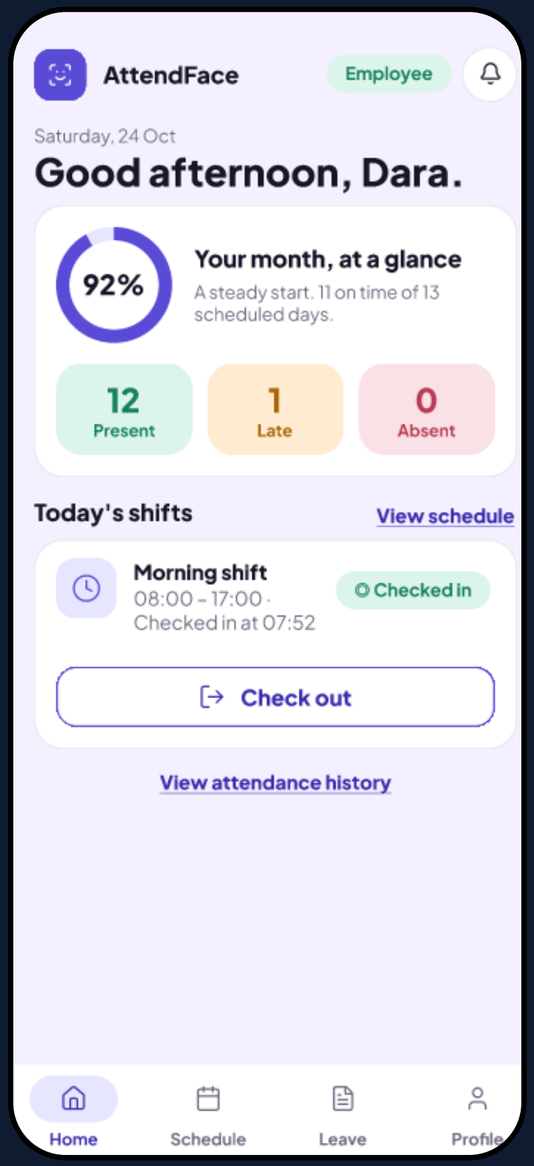
OUR SOLUTION

FaceScan automates attendance with face recognition and GPS validation.

01 Recognize the right person

02 Confirm approved location

03 Record attendance securely



System Architecture

Three layers working together for seamless attendance.



Flutter Mobile App

Captures and sends attendance data



FastAPI Backend

Processes requests and manages APIs



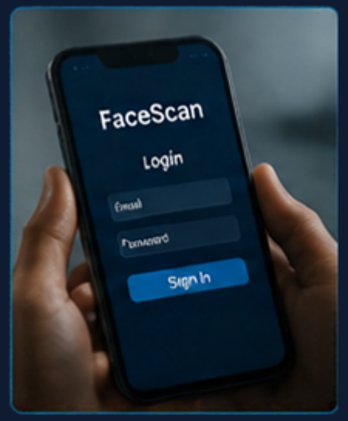
PostgreSQL Database

Stores attendance records securely

Attendance Workflow

From login to attendance saved in five simple steps.

01



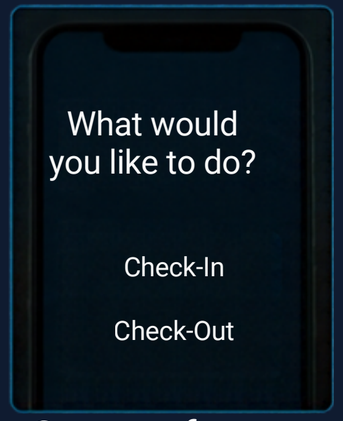
Login / register to FaceTrack

02



Scan user's face

03



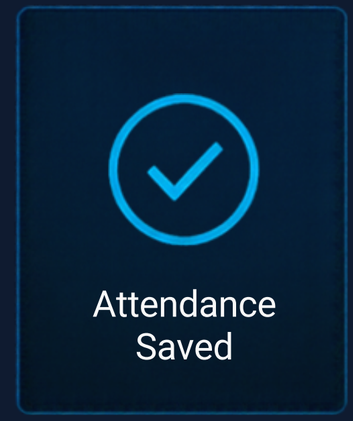
Capture face and GPS

04



Backend validates identity and location

05



Save attendance record

Users and roles

Employee

- Check in / check out
- View own attendance
- Request leave or corrections

Manager

- Review team attendance
- Approve or reject requests
- View team-level records

Admin

- Manage users and settings
- Configure shifts and locations
- View reports and export data

Features overview

Accounts & security

Register, log in, roles, protected routes

Face verification

Face enrollment and template comparison

Attendance

Check-in, check-out, status, timestamps

Leave & corrections

Submit requests and review decisions

Teams & shifts

Groups, schedules, approved locations

Reports & exports

Dashboard, filters, CSV / Excel

How face check-in works

The backend makes the final decision after validating the required checks.

1

Capture

Camera captures a face image.

2

Create template

On-device model creates a 192-number face embedding.

3

Compare

Server compares embeddings using cosine similarity.

4

Validate

Check GPS distance, shift time, and duplicate attendance.

5

Decide

Backend accepts or rejects the check-in.

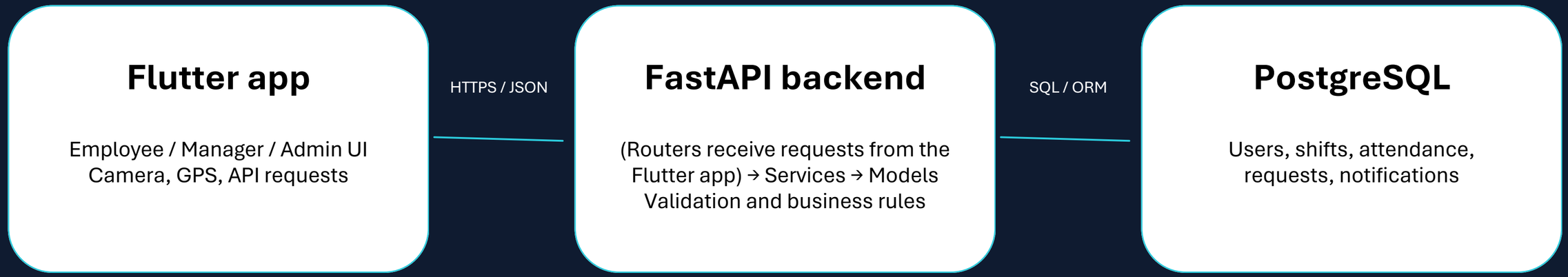
Why the server decides

A modified client app should not be able to skip GPS, schedule, duplicate, or similarity checks. The backend must validate the rules independently.

Confirm that “192 numbers” and the exact comparison flow match your model before presenting.

System architecture

A simple three-part stack, with clear responsibilities in the backend.



Request lifecycle

Flutter sends a request → FastAPI router receives it → service applies rules → model/database stores or retrieves data.

Replace component labels if your final implementation differs.

Check-out and attendance rules

On time	Check-in falls within the allowed shift window.
Late	Check-in occurs after the configured start / grace period.
Absent	No valid check-in is recorded under the attendance policy.
Early leave	Check-out happens before the scheduled end time.
Missing check-out	A check-in exists, but no check-out is recorded.

Corrections and leave

Employee

Submits leave or attendance correction

Manager / Admin

Reviews the request and approves or rejects

System

Updates the relevant record if approved

Notification

Informs the employee of the result

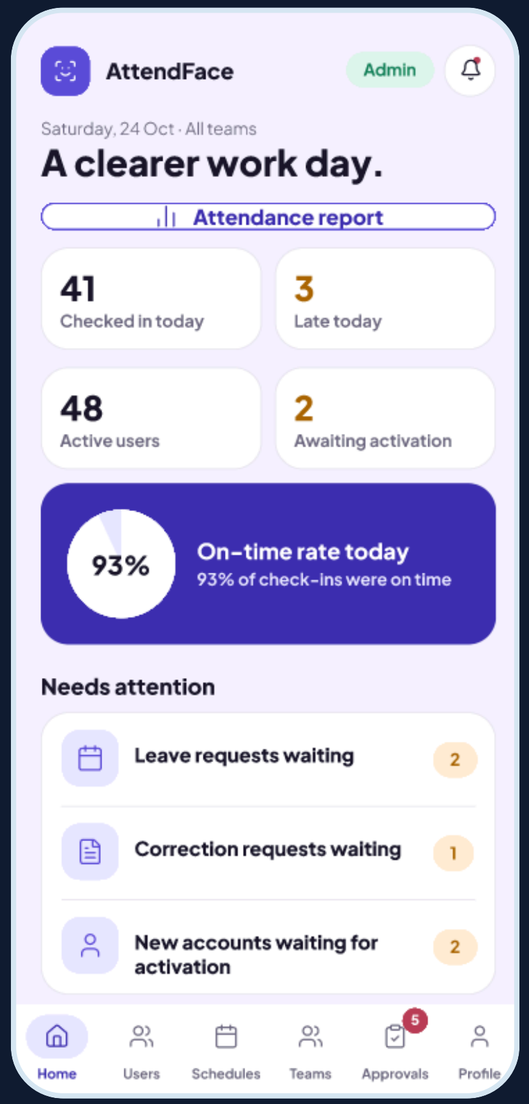
Important

Keep request status, reviewer, and decision time.

Demo opportunity

Show a request moving from Pending to Approved.

Admin tools



- **User management**

  Create, update, activate, or review accounts

- **Teams & shifts**

  Organize groups and working schedules

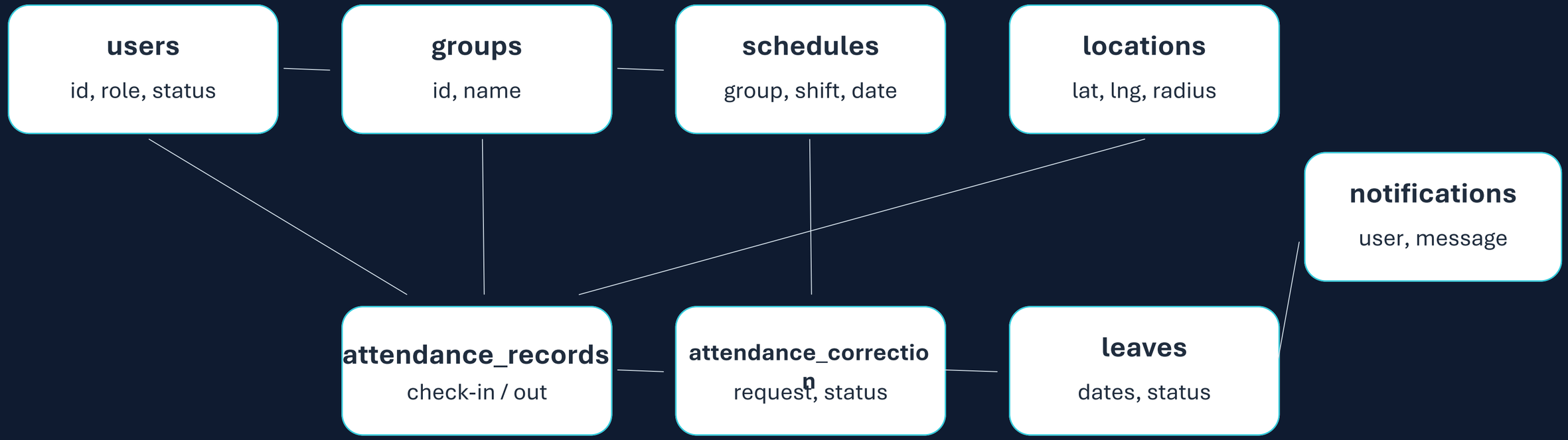
- **Locations**

  Set approved attendance locations and radius

- **Reports & exports**

  Filter records and export CSV / Excel

Database design



Security and validation

Password hashing

Store a password hash, never a plain-text password.

JWT authentication

Use a signed token to identify an authenticated user.

Role-based access

Protect endpoints according to employee, manager, and admin permissions.

Input validation

Validate request data and reject invalid or inconsistent values.

Privacy

If the system stores templates rather than photos, explain what is stored and why.

Live demo

Keep the demo short, predictable, and focused on one end-to-end workflow.

01

Employee

Sign in and check in using face + GPS.

02

Manager

Open a request and approve or reject it.

03

Admin

View attendance and export a report.



That's the end
of our presentation